DURATION	EVENTS DELAYED	EVENTS LOST	CUSTOMERS AFFECTED	JIRA
2h 11m	**1.84 M**	**0**	**~24,000**	**INC-5044**
13:04 - 15:15 UTC, 2026-07-09	peak consumer lag 1,842,000	all replayed from the log	delayed confirmation email	actions STR-902 to STR-906

Description

On 9 July 2026 the `order-events` consumer group stopped making progress at 13:04 UTC. Producers continued to write normally, so no events were lost, but every downstream effect of an order — confirmation email, warehouse pick request, loyalty accrual, fraud scoring — was delayed by up to 2 hours 11 minutes. Consumer lag grew steadily to 1.84 million events before recovery. Order placement itself, which is synchronous, was never affected: customers could check out normally and saw nothing wrong until their confirmation email failed to arrive.

Root cause

> The 8 July upgrade of the Kafka client library from 3.4.1 to 3.7.0 changed the default `max.poll.interval.ms` from 300,000 to 60,000 ms. The enrichment step in this consumer routinely takes 70–90 seconds per batch when the pricing service is slow. Under the new default, the broker judged the consumer dead mid-batch and triggered a rebalance; the rebalance revoked the partitions the consumer was still working on, which caused the next poll to fail and start another rebalance. The group entered a continuous rebalance loop and committed no offsets for two hours.

The upgrade was a transitive bump: the application pinned an internal SDK, and the SDK's minor release moved the Kafka client. No release note in our own changelog mentioned it.

Timeline (UTC)

Time	Event
07-08 16:20	internal-sdk 9.4.0 deployed; pulls kafka-clients 3.7.0.
07-09 13:04	First rebalance. Offset commits stop.
13:19	Lag alert fires at 100k events (warning, no page).
13:52	Lag crosses 500k. Paging alert fires. p.almeida acknowledges.
14:05	SEV-2 declared. Rebalance loop visible in the group coordinator log.
14:31	Diff of the SDK bump reviewed; the changed default is found.
14:48	max.poll.interval.ms pinned to 300000 in the consumer config; change approved.
15:02	Rolling restart of the consumer group begins.
15:15	Group stable, offsets committing, lag draining at 42k events/min.
16:03	Lag back to zero. Incident closed.
07-16	Postmortem review.

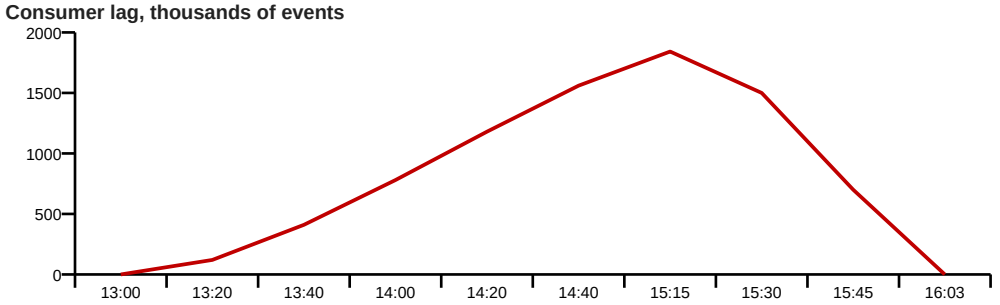
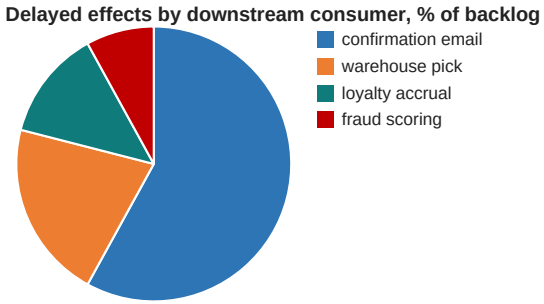


Figure 1 — lag peaks at mitigation and drains in 48 minutes.



Action plan

Jira	Action	Owner	Due	St.
STR-902	Pin every Kafka consumer tuning value explicitly in config; never rely on client defaults.	p.almeida	2026-07-22	Done
STR-903	Fail the build when a transitive dependency changes a documented default; publish a diff of effective config on every release.	d.fischer	2026-08-05	Done
STR-904	Page on consumer lag at 100k rather than warn, and alert separately on 'no offset commit for 5 minutes'.	p.almeida	2026-07-24	Done
STR-905	Move enrichment off the poll thread so a slow pricing call cannot trigger a rebalance.	k.oyelaran	2026-08-19	Done
STR-906	Add a rebalance-storm runbook with the three commands used here.	p.almeida	2026-07-20	Done